Position, distance & displacement

Distance = add all path lengths.

Displacement = final position - initial position.

Math Skills

W1a. 12 + 5 **W1b.** 9 - (-6) **W1c.** 18 - 18

Practice

P01. The hallway delivery

A student carries a box 12 m east from the classroom to the office, then walks 5 m west to a supply closet. What total distance does the student travel?

P02. Across the reference line

On a straight east-west trail, the water fountain is position 0 m. A walker starts at -6 m and finishes at +9 m. Take east as positive. Find the displacement, including direction.

P03. One complete lap

A runner completes one full lap of a circular track and stops exactly where she started. Is her distance zero, her displacement zero, both, or neither? Explain using the start and finish.

P04. Seated on the train

A passenger stays seated while a train moves east at 10 m/s. Describe the passenger's motion relative to the train, and then relative to the station platform. Why do the descriptions differ?

Speed & velocity

Average speed = total distance ÷ total elapsed time.

Distance = speed × time. Velocity is speed with direction.

Math Skills

W2a. 180 ÷ 120

W2b. 4 × 45

W2c. Solve: 3t = 90.

Practice

P05. Walking the trail

A student walks 180 m along a straight trail in 120 s. What is the student's average speed? Include the correct unit.

P06. Cycling along the path

A cyclist rides straight east at a steady 4 m/s for 45 s. How far does the cyclist travel?

P07. Jogging in opposite directions

Two joggers pass a fountain. One moves east at 2 m/s; the other moves west at 2 m/s. Do they have the same speed? Do they have the same velocity? Explain.

P08. The dog stops to sniff

A dog travels along a path at 2 m/s for 10 s, stops for 10 s, then continues at 2 m/s for 10 s. Is its average speed for all 30 s equal to 2 m/s? Explain why the stop matters.

Acceleration

a = $(v_f - v_i) \div \Delta t$ $\mathbf{v_f} = v_i + a \times \Delta t$ **Δt** = final time - initial time

v_i = initial velocity; v_f = final velocity. Right/east is positive.

Math Skills

W3a. 8 - 3

W3b. (14 - 4) ÷ 5

W3c. 6 + (-0.8 × 5)

Practice

P09. The bus pulls away

A bus travels east at 4 m/s when a stopwatch reads 3 s. At 8 s, it travels east at 14 m/s. What is its average acceleration during this interval?

P10. Braking on a scooter

A scooter moves east at 6 m/s. The rider brakes with an average acceleration of -0.8 m/s² for 5 s, while still moving east. What is the final velocity?

P11. Backing up faster

A small remote-control car moves west. Its velocity changes from -1 m/s to -3 m/s in 2 s. Is it speeding up or slowing down? What does the negative sign tell you?

P12. Turning without speeding up

A cyclist follows a circular path at a steady 3 m/s. Is the cyclist accelerating even though the speed stays the same? Explain what is changing.

Reading motion graphs

Slope = change in vertical value ÷ change in time.

Distance-time slope gives average speed for that interval. For straight motion in the positive direction, speed-time slope gives acceleration.

Math Skills

W4a. (30 - 0) ÷ (30 - 0)

W4b. (8 - 2) ÷ (6 - 0)

W4c. Compare: 8/10, 16/10, 4/10. Which is greatest?

Practice

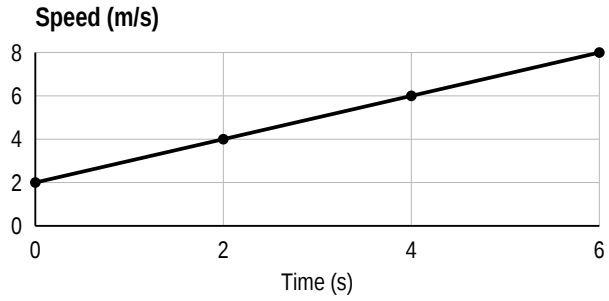
P13. A pause on the nature walk

The table shows a student's total distance along a trail. The student walks forward, pauses, then walks forward again. What is the average speed over the entire 30 s, including the pause?

Time (s)	0	10	20	30
Total distance (m)	0	12	12	30

P14. Reading a scooter's speed graph

The scooter's speed increases steadily as shown. It travels east throughout. Use two points to find its acceleration from 0 s to 6 s.



P15. Which interval is fastest?

A walker's distance-time data are shown below. Which 10 s interval has the greatest average speed? Explain how the distances, or the steepness of a graph, support your choice.

Time (s)	0	10	20	30
Total distance (m)	0	8	24	28

P16. What a speed graph cannot tell you

Two cyclists each ride steadily at 3 m/s for 6 s. One rides east and one rides west. Would their speed-time graphs differ? Could those graphs alone tell you their directions?

Gravity & friction

Weight = mass × g. Use g = 10 m/s² on Earth.

At steady velocity, $\mathbf{F_{net} = 0}$. Static friction adjusts up to its maximum.

Math Skills

W5a. 2.5 × 10

W5b. Solve: f - 35 = 0.

W5c. 1.6 × 150

Practice

P17. How heavy is the backpack?

A backpack has a mass of 2.5 kg. What is its weight near Earth's surface? Use g = 10 m/s².

P18. Keeping the box moving steadily

A student pushes a plastic equipment box across a smooth floor at a steady speed. The push is 35 N to the right. What is the sliding friction force on the box? Give its size and direction.

P19. The same astronaut on the Moon

An astronaut and suit have a combined mass of 150 kg. Compare their mass and weight on Earth and the Moon, where gravity is weaker. Which changes, and why?

P20. A crate that will not move

A student pulls a crate right with 10 N, but it stays at rest. Static friction can reach a maximum of 18 N. What is the actual friction force now? Explain why it is not automatically 18 N.

Net force & Newton's first law

$\mathbf{F_{net}}$ = rightward forces - leftward forces. Choose right as positive.
$\mathbf{F_{net} = 0}$ means constant velocity, which can include rest.

Math Skills

W6a. 45 - 20 **W6b.** 18 + (-30) **W6c.** 18 - 18

Practice

P21. Pulling the supply cart

A student pulls a supply cart right with 45 N. The total friction force is 20 N left. Find the horizontal net force on the cart.

P22. Two ropes, one cart

Two students pull a low-friction cart in opposite directions. One pulls 18 N right; the other pulls 30 N left. Take right as positive. What is the net force? Give its sign and direction.

P23. Moving with balanced forces

A cart rolls right at a steady 1.2 m/s. The pull is 18 N right and friction is 18 N left. Must the cart stop because the forces balance? Explain what zero net force actually means.

P24. After the hockey stick lets go

A hockey puck glides across very smooth ice after the stick is no longer touching it. Over a short interval, ignore the small friction and air resistance. What happens to its horizontal velocity? Is a continuing forward force required?

Newton's second law

$F_{net} = ma$ **$a = F_{net} \div m$** **$m = F_{net} \div a$**
Force: N Mass: kg Acceleration: m/s²

Math Skills

W7a. 50 ÷ 25

W7b. Solve: 0.6m = 18.

W7c. 24 ÷ 12; 24 ÷ 24

Practice

P25. Accelerating the equipment cart

A 25 kg equipment cart has a horizontal net force of 50 N to the right. What is its acceleration?

P26. Finding the trolley's mass

A loaded trolley accelerates right at 0.6 m/s² when its horizontal net force is 18 N right. What is the trolley's mass, including its load?

P27. Empty cart or loaded cart?

The same 24 N net force acts on a 12 kg cart and a 24 kg cart. Both start at rest. Which accelerates more, and how does doubling the mass affect the acceleration?

P28. Force through a bicycle turn

A cyclist makes a circular turn on level pavement at constant speed. In which direction must the net horizontal force point: forward, outward, or toward the center? Which contact force from the road can provide it?

Newton's third law & momentum

Momentum: p = mv. For carts that stick: $\mathbf{v_f = (m_1v_1 + m_2v_2) \div (m_1 + m_2)}$.

Use this collision model when external forces have negligible effect during collision.

A third-law force pair is equal and opposite and acts on two different objects.

Math Skills

W8a. 6 × 3

W8b. 0.5 × 1.2

W8c. 0.6 ÷ (0.5 + 0.5)

Practice

P29. The bowling ball

A 6 kg bowling ball rolls down the lane at 3 m/s. What is its forward momentum?

P30. Carts that latch together

On a level, low-friction track, a 0.5 kg cart moving right at 1.2 m/s hits a stationary 0.5 kg cart. They latch together. Ignore external horizontal forces during the brief collision. What is their shared velocity immediately afterward?

P31. Skaters push apart

Two skaters, one 40 kg and one 60 kg, push off each other on smooth ice. During contact, does either exert a larger force on the other? Do equal force sizes mean equal accelerations? Explain.

P32. Pushing off the pool wall

A swimmer pushes backward on the pool wall. The wall pushes forward on the swimmer. Why do these equal and opposite forces not cancel when you find the net force on the swimmer?